

The Lévy LIBOR model

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Abstract. Models driven by Lévy processes are attractive because of their greater flexibility compared to classical diffusion models. First we derive the dynamics of the LIBOR rate process in a semimartingale as well as a Lévy Heath-Jarrow-Morton setting. Then we introduce a Lévy LIBOR market model. In order to guarantee positive rates, the LIBOR rate process is constructed as an ordinary exponential. Via backward induction we get that the rates are martingales under the corresponding forward measures. An explicit formula to price caps and floors which uses bilateral Laplace transforms is derived.

Key words: Lévy processes, LIBOR market model, forward process, instantaneous forward rates, caps, floors

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1 Introduction

The preferred approach to model interest rate term structures, the Heath-Jarrow-Morton framework, assumes continuously compounding rates. In real markets however the interest on invested money typically accrues according to a discrete time grid, the so-called tenor structure. Sandmann et al. (1995) were the first who tried to reconcile these two diverging views. The approach became popular under the

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name LIBOR market models when Miltersen et al. (1997), Brace et al. (1997), and Jamshidian (1997) succeeded to construct arbitrage-free term structure models which are consistent with the market practice of pricing caps and floors by Black's (1976) formula. Black's formula for caps is motivated by the Black-Scholes (1973) option pricing formula and assumes that the LIBOR rates follow a geometric Brownian motion. The purpose of this paper is to study a more general approach where LIBOR rates are driven by a Lévy process instead of a Brownian motion. An introduction to LIBOR models can be found in several recent books, see e.g. Musiela and Rutkowski (1998), Brigo and Mercurio (2001), or Rutkowski (2001).

This article is related to Jamshidian (1999) where a LIBOR rate process is studied which is driven by a general semimartingale, but pricing of caps and floors is not considered. Thus this paper can be seen in parts as a special case of Jamshidian's approach, but we specify the driving Lévy process and show how the LIBOR rates can be represented as ordinary exponentials of stochastic integrals driven by a (non-homogeneous) Lévy process. This will guarantee nonnegative rates. Furthermore we derive an explicit pricing formula for caps which uses bilateral Laplace transforms. The latter is based on an idea developed in Raible (2000).

The Lévy term structure approach can be embedded in Jamshidian's general semimartingale setting as has been shown in Özkan (2002). For a given discrete-tenor structure Jamshidian (1999) models the bond market by a finite number of strictly positive semimartingales which are not specified any further. Even negative LIBOR rates can occur. Furthermore, within this bond market, there may not be a risk-neutral measure $\mathbb{P}^*$. It exists if a specific process, the state-price density, is a special semimartingale. For details we refer to Sect. 4 in Jamshidian (1999). Our aim in this paper is not to develop an abstract LIBOR market theory which is as general as possible. Our interest is to push further the market practice to price caps and floors by Black's formula to the derivation of a formula which allows for more statistical accuracy. The natural candidates for driving Lévy processes in this context are generalized hyperbolic Lévy motions since they guarantee high flexibility based on five parameters only. In addition for those processes the Gaussian part is zero, which simplifies their implementation considerably.

2 Modeling instantaneous forward rates

LIBOR rates can be derived from bond prices which are determined by instantaneous forward rates. The advantage of this approach is that LIBOR rates can be embedded in an existing HJM-framework as it was done in Brace et al. (1997). We study this approach in the general setting of Björk et al. (1997) and in the Lévy setting of Eberlein and Raible (1999), Eberlein and Özkan (2003), and Eberlein et al. (2003).

2.1 The semimartingale setting

We assume a complete stochastic basis $(\Omega, \mathcal{F}_{T^*}, \mathbb{P}^*, (\mathcal{F}_t)_{0 \leq t \leq T^*})$, where the filtration satisfies the usual conditions. The predictable σ -field is denoted by $\mathcal{P}$. For further notation we refer the reader to Jacod and Shiryaev (1987).

Our starting point is the following assumption on the dynamics of the instantaneous forward rates (see Björk et al. 1997, Assumption 5.1).

Assumption (S). The dynamics of the instantaneous forward rates for $T \in [0, T^*]$ is given by

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T)^\top dW_t^* + \int_{\mathbb{R}^r} \delta(t, x, T)(\mu - \nu^*)(dt, dx), \quad (2.1)$$

where W^* is a standard Brownian motion in $\mathbb{R}^d$, μ is the random measure of jumps of a semimartingale with continuous compensator ν^* , i.e. there exists $\lambda^* : \mathbb{R}_+ \times \mathcal{B}(\mathbb{R}^r) \rightarrow \mathbb{R}_+$ such that $\nu^*(dt, dx) = \lambda^*(t, dx) dt$. The coefficients are continuous in the second variable, where $\alpha : \Omega \times [0, T^*] \times [0, T^*] \rightarrow \mathbb{R}$ and $\sigma : \Omega \times [0, T^*] \times [0, T^*] \rightarrow \mathbb{R}_+^d$ are $\mathcal{P} \otimes \mathcal{B}([0, T^*])$ -measurable and $\delta : \Omega \times [0, T^*] \times \mathbb{R}^r \times [0, T^*] \rightarrow \mathbb{R}$ is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}^r) \otimes \mathcal{B}([0, T^*])$ -measurable. The coefficients satisfy the following conditions. Denote $\Delta := \{(s, u) \in \mathbb{R}_+ \times \mathbb{R}_+ \mid 0 \leq s \leq u \leq T^*\}$. Then if $(t, T) \notin \Delta$, we have $\alpha(t, T) = \delta(t, x, T) = 0$, $\sigma(t, T) = (0, \dots, 0)^\top$, and for all $(t, T) \in \Delta$

$$\begin{aligned} \int_0^T \int_t^T |\alpha(s, u)| du ds &< \infty, & \int_0^T \int_t^T |\sigma(s, u)|^2 du ds &< \infty, \\ \int_0^T \int_{\mathbb{R}^r} \int_t^T |\delta(s, x, u)|^2 du \nu^*(ds, dx) &< \infty. \end{aligned}$$

We put

$$\begin{aligned} A(t, T) &:= - \int_t^T \alpha(t, u) du, & S(t, T) &:= - \int_t^T \sigma(t, u) du, \\ \text{and } D(t, x, T) &:= - \int_t^T \delta(t, x, u) du. \end{aligned}$$

Then $\mathbb{P}^*$ is a martingale measure or we could say the model is arbitrage-free iff

$$\int_0^t \int_{\mathbb{R}^r} \left(e^{D(s, x, T)} - 1 - D(s, x, T) \right) \nu^*(ds, dx) < \infty \quad \text{for any } (t, T) \in \Delta$$

and

$$A(t, T) + \frac{1}{2} |S(t, T)|^2 + \int_{\mathbb{R}^r} \left(e^{D(t, x, T)} - 1 - D(t, x, T) \right) \lambda^*(t, dx) = 0 \quad (2.2)$$

$d\mathbb{P}^* \otimes dt$ -a.e.

(see Björk et al. 1997, Proposition 5.3).

Given the forward rates $f(t, T)$ the price at time t of a zero-coupon bond with maturity T is defined by $B(t, T) = \exp \left(- \int_t^T f(t, u) du \right)$. From (2.1) one easily

derives

$$B(t, T) = B(0, T) \exp \left(\int_0^t (f(s, s) + A(s, T)) ds + \int_0^t S(s, T)^\top dW_s^* \right. \\ \left. + \int_0^t \int_{\mathbb{R}^r} D(s, x, T) (\mu - \nu^*)(ds, dx) \right). \quad (2.3)$$

Remember that the savings account is $B_t = \exp \left(\int_0^t f(s, s) ds \right)$. Denote the discounted bond price at time t by

$$Z(t, T) := \frac{B(t, T)}{B_t}.$$

In the risk-neutral case, i.e. under (2.2), the price formula (2.3) translates into

$$Z(t, T) = Z(0, T) \exp \left(\int_0^t S(s, T)^\top dW_s^* - \frac{1}{2} \int_0^t |S(s, T)|^2 ds \right. \\ \left. + \int_0^t \int_{\mathbb{R}^r} D(s, x, T) (\mu - \nu^*)(ds, dx) \right. \\ \left. - \int_0^t \int_{\mathbb{R}^r} \left(e^{D(s, x, T)} - 1 - D(s, x, T) \right) \lambda^*(s, dx) ds \right). \quad (2.4)$$

For two maturities T, U the forward process is defined by

$$F_B(t, T, U) := \frac{B(t, T)}{B(t, U)}. \quad (2.5)$$

Let $\delta > 0$ with $\delta \leq T^* - T$. Then the δ -(forward) -LIBOR rates are defined by

$$L(t, T) = \frac{1}{\delta} (F_B(t, T, T + \delta) - 1) \quad (2.6)$$

$$= \frac{1}{\delta} \left(\frac{B(t, T)}{B(t, T + \delta)} - 1 \right). \quad (2.7)$$

The δ -(forward)-LIBOR rate coincides with the forward swap rate of a single-period swap settled in arrears (see e.g. Rutkowski 2001). When we speak of LIBOR rates, we always mean forward LIBOR rates. The next theorem states the dynamics of $L(., T)$ under the risk-neutral measure $\mathbb{P}^*$.

Theorem 2.1 *If $f(., T)$ satisfies (2.1) and $A(., T)$, $D(., ., T)$, and $S(., T)$ satisfy (2.2), then the dynamics of $L(., T)$ under $\mathbb{P}^*$ is given by*

$$\frac{\delta}{1 + \delta L(t, T)} dL(t, T) = (S(t, T + \delta) - S(t, T))^\top S(t, T + \delta) dt \\ + \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T + \delta)} - 1 + e^{D(t, x, T + \delta)} - e^{D(t, x, T)} \right) \nu^*(dt, dx) \\ + (S(t, T) - S(t, T + \delta))^\top dW_t^* \quad (2.8) \\ + \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T + \delta)} - 1 \right) (\mu - \nu^*)(dt, dx).$$

Proof From (2.3) we get

$$\begin{aligned} F_B(t, T, T + \delta) &= \frac{B(0, T)}{B(0, T + \delta)} \exp \left(\int_0^t (A(s, T) - A(s, T + \delta)) ds \right. \\ &\quad + \int_0^t (S(s, T) - S(s, T + \delta))^\top dW_s^* \\ &\quad \left. + \int_0^t \int_{\mathbb{R}^r} (D(s, x, T) - D(s, x, T + \delta))(\mu - \nu^*)(ds, dx) \right). \end{aligned}$$

The dynamics of the LIBOR rate is related to the dynamics of the forward process in the following way

$$\frac{\delta}{1 + \delta L(t-, T)} dL(t, T) = \frac{1}{F_B(t-, T, T + \delta)} dF_B(t, T, T + \delta).$$

Since $F_B(\cdot, T, T + \delta)$ is a positive semimartingale, it can be expressed as the stochastic exponential of the stochastic logarithm $\mathcal{L}(F_B(\cdot, T, T + \delta))$. The stochastic logarithm which is defined by $\int \frac{1}{F_B-} dF_B$ can be represented as follows (see Kallsen and Shiryaev 2002, Lemma 2.4):

$$\begin{aligned} \mathcal{L}(F_B(\cdot, T, T + \delta))_t & \quad (2.9) \\ &= \ln \left(\frac{F_B(t, T, T + \delta)}{F_B(0, T, T + \delta)} \right) + \int_0^t \frac{1}{2F_B^2(s-, T, T + \delta)} d \langle F_B^c(\cdot, T, T + \delta) \rangle_s \\ &\quad - \int_0^t \int_{\mathbb{R}^r} \left(\ln \left(1 + \frac{x}{F_B(s-, T, T + \delta)} \right) - \frac{x}{F_B(s-, T, T + \delta)} \right) \mu^{F_B}(ds, dx). \end{aligned}$$

The first expression on the right-hand side is simply

$$\begin{aligned} &\int_0^t (A(s, T) - A(s, T + \delta)) ds + \int_0^t (S(s, T) - S(s, T + \delta))^\top dW_s^* \\ &\quad + \int_0^t \int_{\mathbb{R}^r} (D(s, x, T) - D(s, x, T + \delta))(\mu - \nu^*)(ds, dx). \end{aligned} \quad (2.10)$$

For the next expression on the right-hand side of Eq. (2.9) we have to calculate the predictable quadratic variation (or Meyer process) of $F_B(\cdot, T, T + \delta)$. It can be derived from the Meyer process of $\ln F_B(\cdot, T, T + \delta)$. Note that

$$\langle (\ln F_B(\cdot, T, T + \delta))^c \rangle_t = \int_0^t |S(s, T) - S(s, T + \delta)|^2 ds.$$

Therefore, the predictable quadratic variation of $F_B(\cdot, T, T + \delta)$ is

$$\langle (F_B(\cdot, T, T + \delta))^c \rangle_t = \int_0^t F_B^2(s-, T, T + \delta) |S(s, T) - S(s, T + \delta)|^2 ds.$$

Hence, the second expression on the right-hand side of (2.9) simplifies to

$$\int_0^t \frac{1}{2} |S(s, T) - S(s, T + \delta)|^2 ds. \quad (2.11)$$

For the remaining expression on the right-hand side of (2.9), we have to determine μ^{F_B} , the random measure of jumps of F_B . The jumps of F_B and $\ln F_B$ are related by

$$\Delta F_B(t, T, T + \delta) = F_B(t-, T, T + \delta) \left(e^{\Delta \ln F_B(t, T, T + \delta)} - 1 \right).$$

As one can see from (2.10) the jumps of $\ln F_B(\cdot, T, T + \delta)$ are given by

$$\Delta \ln F_B(t, T, T + \delta) = \int_{\mathbb{R}^r} (D(t, x, T) - D(t, x, T + \delta)) \mu(\{t\}, dx).$$

Therefore for μ^{F_B} -integrable functions g the following equation holds:

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}^r} g(s, y) \mu^{F_B}(ds, dy) \\ &= \int_0^t \int_{\mathbb{R}^r} g(s, F_B(s-, T, T + \delta)(e^x - 1)) \mu^{\ln F_B}(ds, dx) \\ &= \int_0^t \int_{\mathbb{R}^r} g(s, F_B(s-, T, T + \delta)(\exp(D(s, x, T) - D(s, x, T + \delta)) - 1)) \mu(ds, dx). \end{aligned}$$

Hence,

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}^r} \left(\ln \left(\left| 1 + \frac{x}{F_B(s-, T, T + \delta)} \right| \right) - \frac{x}{F_B(s-, T, T + \delta)} \right) \mu^{F_B}(ds, dx) \quad (2.12) \\ &= \int_0^t \int_{\mathbb{R}^r} \left((D(s, x, T) - D(s, x, T + \delta)) - e^{D(s, x, T) - D(s, x, T + \delta)} + 1 \right) \mu(ds, dx). \end{aligned}$$

Now we insert (2.10) – (2.12) in (2.9) and get the dynamics of $F_B(\cdot, T, T + \delta)$

$$\begin{aligned} \frac{dF_B(t, T, T + \delta)}{F_B(t-, T, T + \delta)} &= (A(t, T) - A(t, T + \delta)) dt + (S(t, T) - S(t, T + \delta))^\top dW_t^* \\ &+ \int_{\mathbb{R}^r} (D(t, x, T) - D(t, x, T + \delta)) (\mu - \nu^*)(dt, dx) \\ &+ \frac{1}{2} |S(t, T) - S(t, T + \delta)|^2 dt \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T + \delta)} - 1 - (D(t, x, T) - D(t, x, T + \delta)) \right) \mu(dt, dx) \\ &= \left(A(t, T) - A(t, T + \delta) + \frac{1}{2} |S(t, T) - S(t, T + \delta)|^2 \right) dt \\ &+ (S(t, T) - S(t, T + \delta))^\top dW_t^* \\ &+ \int_{\mathbb{R}^r} (D(t, x, T) - D(t, x, T + \delta)) (\mu - \nu^*)(dt, dx) \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T + \delta)} - 1 - (D(t, x, T) - D(t, x, T + \delta)) \right) (\mu - \nu^*)(dt, dx) \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T + \delta)} - 1 - (D(t, x, T) - D(t, x, T + \delta)) \right) \nu^*(dt, dx). \end{aligned}$$

Under the martingale assumption (2.2) we get

$$\begin{aligned} \frac{dF_B(t, T, T+\delta)}{F_B(t-, T, T+\delta)} &= \frac{1}{2} \left(|S(t, T+\delta)|^2 - |S(t, T)|^2 + |S(t, T) - S(t, T+\delta)|^2 \right) dt \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T+\delta)} + e^{D(t, x, T+\delta)} - e^{D(t, x, T)} - 1 \right) \nu^*(dt, dx) \\ &+ (S(t, T) - S(t, T+\delta))^\top dW_t^* \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, T+\delta)} - 1 \right) (\mu - \nu^*)(dt, dx). \end{aligned}$$

Note that

$$\begin{aligned} &|S(t, T+\delta)|^2 - |S(t, T)|^2 + |S(t, T) - S(t, T+\delta)|^2 \\ &= 2(S(t, T+\delta) - S(t, T))^\top S(t, T+\delta). \end{aligned}$$

This proves the theorem. $\square$

Given bond prices $B(t, T)$ and the related savings account B_t under the martingale measure $\mathbb{P}^*$, we define for any $T \leq T^*$ the forward measure $\mathbb{P}_T$ associated with the settlement date T by the formula

$$\frac{d\mathbb{P}_T}{d\mathbb{P}^*} = \frac{1}{B_T B(0, T)} \quad \mathbb{P}^*\text{-a.s.} \quad (2.13)$$

Note that due to the martingale property of $B(t, T)/B_t$ this Radon-Nikodym derivative, when restricted to $\mathcal{F}_t$, is given by

$$\eta_t = \mathbb{E}_{\mathbb{P}^*} \left[\frac{1}{B_T B(0, T)} \mid \mathcal{F}_t \right] = \frac{B(t, T)}{B_t B(0, T)}.$$

When the bond price is of the form (2.3) we get in the risk-neutral case

$$\begin{aligned} \eta_t &= \exp \left(\int_0^t S(s, T)^\top dW_s^* - \frac{1}{2} \int_0^t |S(s, T)|^2 ds \right. \\ &+ \int_0^t \int_{\mathbb{R}^r} D(s, x, T) (\mu - \nu^*)(ds, dx) \\ &\left. - \int_0^t \int_{\mathbb{R}^r} \left(e^{D(s, x, T)} - 1 - D(s, x, T) \right) \nu^*(ds, dx) \right). \end{aligned} \quad (2.14)$$

Alternatively the forward measures $\mathbb{P}_U$ for $U \leq T^*$ can be introduced by requiring that for any $T \in [0, U]$ the forward process $F_B(t, T, U) = B(t, T)/B(t, U)$ is a $\mathbb{P}_U$ -martingale. The martingale property can also be seen from the following proposition.

Proposition 2.2 *The $\mathbb{P}_U$ -dynamics of $F_B(\cdot, T, U)$ is given by*

$$\begin{aligned} dF_B(t, T, U) &= F_B(t-, T, U) \left((S(t, T) - S(t, U))^\top dW_t^U \right. \\ &\left. + \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 \right) (\mu - \nu^U)(dt, dx) \right) \end{aligned}$$

where $W_t^U = W_t^* - \int_0^t S(s, U)ds$ and ν^U , the $\mathbb{P}_U$ -compensator of μ , is given by

$$\nu^U(ds, dx) = \exp(D(s, x, U))\nu^*(ds, dx).$$

Proof In exactly the same way as in the proof of Theorem 2.1 – replace $T + \delta$ by U – we can derive the dynamics of the forward process under $\mathbb{P}^*$ as

$$\begin{aligned} & \frac{1}{F_B(t-, T, U)} dF_B(t, T, U) \\ &= (S(t, U) - S(t, T))^{\top} S(t, U)dt + (S(t, T) - S(t, U))^{\top} dW_t^* \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 \right) (\mu - \nu^*)(dt, dx) \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 + e^{D(t, x, U)} - e^{D(t, x, T)} \right) \nu^*(dt, dx). \end{aligned}$$

The first two expressions on the right-hand side of the last equation can be written as

$$(S(t, T) - S(t, U))^{\top} (dW_t^* - S(t, U)dt).$$

Now we look at the terms related to the jumps of the forward process

$$\begin{aligned} & \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 \right) (\mu - \nu^*)(dt, dx) \\ &+ \int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 \right) \left(1 - e^{D(t, x, U)} \right) \nu^*(dt, dx). \end{aligned}$$

Since $(1 - e^{D(t, x, U)}) \nu^*(dt, dx) = (\nu^* - \nu^U)(dt, dx)$ the sum of the two integrals is

$$\int_{\mathbb{R}^r} \left(e^{D(t, x, T) - D(t, x, U)} - 1 \right) (\mu - \nu^U)(dt, dx). \quad \square$$

2.2 The Lévy setting

For implementation the semimartingale setting is too general. Therefore we briefly outline the Lévy setting which was developed in increasing generality in Eberlein and Raible (1999), Eberlein and Özkan (2003), and Eberlein et al. (2003). This approach is only slightly more restrictive but can be implemented based on a few parameters only. For more details see the references above. To simplify notation we concentrate here on the case of a one-dimensional driving process.

Assumption ($\mathbb{L}$). The dynamics of the instantaneous forward rates for $T \in [0, T^*]$ is given by

$$df(t, T) = \alpha(t, T)dt + \sigma(t, T)dL_t \quad (2.15)$$

where

$$L_t = \int_0^t b_s ds + \int_0^t c_s^{1/2} dW_s + \int_0^t \int_{\mathbb{R}} x(\mu - \nu)(ds, dx) \quad (2.16)$$

is a non-homogeneous Lévy process defined by its (deterministic) characteristics b_s, c_s, F_s , where $\nu(ds, dx) = F_s(dx)ds$. Those characteristics have to satisfy certain integrability conditions: $\int_0^{T^*} (|b_s| + |c_s| + \int_{\{|x| \leq 1\}} x^2 F_s(dx))ds < \infty$ and there is a constant M such that for $|u| \leq M$, $\int_0^{T^*} \int_{\{|x| > 1\}} \exp(ux) F_s(dx)ds < \infty$. We assume α, σ to be bounded and for $A(t, T) = -\int_t^T \alpha(t, s)ds$ and $\Sigma(t, T) = -\int_t^T \sigma(t, s)ds$ we assume furthermore that there is a constant $M' < M$ such that $0 \leq \Sigma(t, T) \leq M'$ for all $0 \leq t, T \leq T^*$.

Given forward rates of the form (2.15), the price of a zero-coupon bond with maturity T is derived as

$$B(t, T) = B(0, T)B_t \exp \left(\int_0^t A(s, T)ds + \int_0^t \Sigma(s, T)dL_s \right). \quad (2.17)$$

Discounted bond prices $B(t, T)/B_t$ are martingales if $A(t, T)$ is chosen appropriately, namely

$$A(t, T) = -b_t \Sigma(t, T) - \frac{1}{2} c_t \Sigma(t, T)^2 - \int_{\mathbb{R}} \left(e^{\Sigma(t, T)x} - 1 - \Sigma(t, T)x \right) F_t(dx).$$

It has been shown in Eberlein et al. (2003) that the martingale measure is unique if the driving process (L_t) is 1-dimensional and the coefficients α and σ and thus A and Σ are deterministic. In applications we will always use deterministic volatility structures $\Sigma(t, T)$ such as the Vasiček or the Ho-Lee structure or some more sophisticated ones as given in Brigo and Mercurio (2001). Therefore, we assume that $\Sigma(t, T)$ is deterministic.

Let us describe the effect of the change from the martingale measure to a forward measure $\mathbb{P}_U$. The coefficients $S(s, T)$ and $D(s, x, T)$ from the semi-martingale setting are given in the Lévy setting by $S(s, T) = c_s^{1/2} \Sigma(s, T)$ and $D(s, x, T) = \Sigma(s, T)x$. Thus the change to the forward measure $\mathbb{P}_U$ produces a Brownian motion $W_t^U = W_t - \int_0^t c_s^{1/2} \Sigma(s, U)ds$ and the $\mathbb{P}_U$ -compensator of μ is given by $\nu^U(ds, dx) = \exp(\Sigma(s, U)x)\nu(ds, dx)$. The density applied to ν here depends on s . Therefore, if ν is a product measure $F(dx)ds$ – which is the case for (homogeneous) Lévy processes – ν^U is no longer a product measure. In other words, even if we start with a (homogeneous) Lévy process (L_t) , i.e. b_s, c_s , and F_s do not depend on s , the change to a forward measure $\mathbb{P}_U$ leads to a non-homogeneous Lévy process. This is one of the reasons why we use non-homogeneous Lévy processes from the very beginning. The other reason is a statistical one. With non-homogeneous Lévy processes we gain additional flexibility to fit the data while keeping tractability of the model and the number of parameters to estimate on a reasonable level.

3 A forward price model

Instead of deriving forward processes (or LIBOR rates) from an existing bond price model one can model the forward processes (or LIBOR rates) directly. We

start with the Lévy model for the forward processes. As in the previous section we assume a complete stochastic basis $(\Omega, \mathcal{F}_{T^*}, \mathbb{P}_{T^*}, (\mathcal{F}_t)_{0 \leq t \leq T^*})$ and a family $(B(t, T))_{(t, T) \in \Delta}$ of bond prices to be given. The bond prices are assumed to be positive semimartingales, and we also assume that they satisfy $B(T, T) = 1$ for all $T \in [0, T^*]$. The dynamics of $B(\cdot, T)$ remains unspecified in this section.

To model the forward processes we make the following assumptions.

(FPM.1): For fixed maturity date $T \leq T^*$ the dynamics of the forward process is given by

$$dF_B(t, T, T^*) = F_B(t-, T, T^*) dL_t^{T, T^*},$$

with initial condition

$$F_B(0, T, T^*) = \frac{B(0, T)}{B(0, T^*)},$$

where

$$\begin{aligned} L_t^{T, T^*} &= L_0^{T, T^*} + \int_0^t \alpha(s, T, T^*) dW_s^{T^*} \\ &\quad + \int_0^t \int_{\mathbb{R}} \beta(s, x, T, T^*) (\mu^L - \nu^{T^*, L})(ds, dx), \end{aligned}$$

and α and $\beta > -1$ are deterministic functions such that the integrals are well-defined and with the additional condition that $\alpha(t, T^*, T^*) = \beta(t, x, T^*, T^*) = 0$ for any $t \in [0, T^*]$, and $x \in \mathbb{R}$. Under $\mathbb{P}_{T^*}$, W^{T^*} is a standard Brownian motion, μ^L is the random measure of jumps of a non-homogeneous Lévy process with respect to $\mathbb{P}_{T^*}$ and $\nu^{T^*, L}(dt, dx) = \nu^{T^*}(t, dx)dt$ is the $\mathbb{P}_{T^*}$ -compensator of μ^L which satisfies the following integrability condition: $\int_0^{T^*} \int_{\{|x| > 1\}} \exp(ux) \nu^{T^*}(t, dx)dt < \infty$, for $-M \leq u \leq M$, for some $M > 0$.

(FPM.2): The relationship between the bond price and the forward process is given by

$$F_B(t, T, T^*) = \frac{B(t, T)}{B(t, T^*)}.$$

Assumption (FPM.1) is equivalent to $F_B(\cdot, T, T^*) = F_B(0, T, T^*)\mathcal{E}(L^{T, T^*})$ and hence by Jacod and Shiryaev (1987, I.4.64)

$$\begin{aligned} F_B(t, T, T^*) &= F_B(0, T, T^*) \exp \left(\int_0^t \alpha(s, T, T^*) dW_s^{T^*} - \frac{1}{2} \int_0^t \alpha^2(s, T, T^*) ds \right. \\ &\quad \left. + \int_0^t \int_{\mathbb{R}} \beta(s, x, T, T^*) (\mu^L - \nu^{T^*, L})(ds, dx) \right. \\ &\quad \left. + \int_0^t \int_{\mathbb{R}} (\ln(1 + \beta(s, x, T, T^*)) - \beta(s, x, T, T^*)) \mu^L(ds, dx) \right). \end{aligned} \quad (3.1)$$

By construction $F_B(\cdot, T, T^*)$ is a local $\mathbb{P}_{T^*}$ -martingale. For simplicity we assume that it is a martingale. Under (FPM.2) the martingale property of the forward process is equivalent to

$$B(t, T) = \mathbb{E}_{\mathbb{P}_{T^*}} \left[\frac{B(t, T^*)}{B(T, T^*)} \middle| \mathcal{F}_t \right].$$

From (3.1) we get the forward process $F_B(t, T, U)$ as

$$\begin{aligned} F_B(t, T, U) &= F_B(0, T, U) \exp \left(\int_0^t \alpha(s, T, U) dW_s^{T^*} \right. \\ &\quad - \frac{1}{2} \int_0^t (\alpha^2(s, T, T^*) - \alpha^2(s, U, T^*)) ds \\ &\quad + \int_0^t \int_{\mathbb{R}} \beta(s, x, T, U) (\mu^L - \nu^{T^*, L})(ds, dx) \\ &\quad \left. + \int_0^t \int_{\mathbb{R}} (\ln(\beta'(s, x, T, U)) - \beta(s, x, T, U)) \mu^L(ds, dx) \right), \end{aligned}$$

where for brevity we have set

$$\begin{aligned} \alpha(s, T, U) &:= \alpha(s, T, T^*) - \alpha(s, U, T^*), \\ \beta(s, x, T, U) &:= \beta(s, x, T, T^*) - \beta(s, x, U, T^*), \\ \beta'(s, x, T, U) &:= \frac{1 + \beta(s, x, T, T^*)}{1 + \beta(s, x, U, T^*)}. \end{aligned}$$

By Itô's formula the $\mathbb{P}_{T^*}$ -dynamics of $F_B(\cdot, T, U)$ is

$$\begin{aligned} dF_B(t, T, U) &= F_B(t-, T, U) \left(\alpha(t, T, U) (dW_t^{T^*} - \alpha(t, U, T^*) dt) \right. \\ &\quad + \int_{\mathbb{R}} \beta(t, x, T, U) (\mu^L - \nu^{T^*, L})(dt, dx) \\ &\quad \left. + \int_{\mathbb{R}} (\beta'(t, x, T, U) - \beta(t, x, T, U) - 1) \mu^L(dt, dx) \right). \end{aligned} \quad (3.2)$$

The functions $\phi^U(t) := \alpha(t, U, T^*)$ and $Y^U(t, x) := \beta(t, x, U, T^*) + 1 = \beta'(t, x, U, T^*)$ determine a measure $\mathbb{P}_U \sim \mathbb{P}_{T^*}$ with the property that

$$W_t^U := W_t^{T^*} - \int_0^t \phi^U(s) ds$$

is a standard Brownian motion under $\mathbb{P}_U$ and the $\mathbb{P}_U$ -compensator of μ^L is

$$\nu^{U, L}(dt, dx) := Y^U(t, x) \nu^{T^*, L}(dt, dx).$$

Again the martingality of $F_B(t, T, U)$ becomes visible if its dynamics is described under $\mathbb{P}_U$ because

$$\begin{aligned} dF_B(t, T, U) &= F_B(t-, T, U) \left(\alpha(t, T, U) dW_t^U \right. \\ &\quad \left. + \int_{\mathbb{R}} (\beta'(t, x, T, U) - 1) (\mu^L - \nu^{U, L})(dt, dx) \right). \end{aligned} \quad (3.3)$$

4 The Lévy LIBOR model

In Sects. 2 and 3 we discussed two frameworks where LIBOR rates can be derived from bond prices or from forward processes. In this section the LIBOR rates themselves are subject to modeling. We will show that the market practice of assuming that caps and floors can be priced such that the LIBOR rates are driven by a Wiener process under each forward measure can be pushed further to a model where the LIBOR rates are driven by more general Lévy processes under each forward measure.

We consider a discrete-tenor structure $T_0 < T_1 < T_2 < \dots < T_n < T_{n+1} = T^*$ with $\delta = T_{i+1} - T_i$, $i = 0, \dots, n$ and since we will proceed by a backward induction we write $T_i^* = T^* - i\delta$ for $i = 1, \dots, n$. The explicit construction of the LIBOR rates uses the following inputs.

- (LR.1): For any maturity T_i there is a bounded deterministic function $\lambda(\cdot, T_i)$, which represents the volatility of the forward LIBOR rate process $L(\cdot, T_i)$.
 (LR.2): We assume a strictly decreasing and strictly positive initial term structure $B(0, T)$ ($T \in]0, T^*]$). Consequently the initial term structure $L(0, T)$ of forward LIBOR rates is given for $T \in]0, T^* - \delta]$ by

$$L(0, T) = \frac{1}{\delta} \left(\frac{B(0, T)}{B(0, T + \delta)} - 1 \right).$$

As before we assume a complete stochastic basis $(\Omega, \mathcal{F}_{T^*}, \mathbb{P}_{T^*}, (\mathcal{F}_t)_{0 \leq t \leq T^*})$ to be given. We start the construction by defining the forward LIBOR rate with the longest maturity $L(t, T_1^*)$. Under $\mathbb{P}_{T^*}$ we postulate that

$$L(t, T_1^*) = L(0, T_1^*) \exp \left(\int_0^t \lambda(s, T_1^*) dL_s^{T^*} \right) \quad (4.1)$$

where

$$L_t^{T^*} = \int_0^t b_s ds + \int_0^t c_s^{1/2} dW_s^{T^*} + \int_0^t \int_{\mathbb{R}} x(\mu^L - \nu^{T^*, L})(ds, dx) \quad (4.2)$$

is a non-homogeneous Lévy process, i.e. W^{T^*} is a standard Brownian motion, c_s is positive such that $\int_0^{T^*} (|b_s| + c_s) ds < \infty$, μ^L is the random measure of jumps of the process, and $\nu^{T^*, L}(ds, dx) = F_s(dx)ds$ is the $\mathbb{P}_{T^*}$ -compensator of μ^L . We assume that the Lévy measures F_s , which are measures on $\mathbb{R}$ with $F_s(\{0\}) = 0$ and $\int_0^{T^*} \int_{\mathbb{R}} (x^2 \wedge 1) F_s(dx) ds < \infty$, satisfy the following additional integrability assumption

$$\int_0^{T^*} \int_{\{|x| > 1\}} \exp(ux) F_s(dx) ds < \infty$$

for $|u| \leq (1 + \varepsilon)M$, where $M, \varepsilon > 0$ are constants such that $\sum_{i=1}^n |\lambda(\cdot, T_i^*)| \leq M$. Note that in order to simplify notation we consider a 1-dimensional driving process $(L_t^{T^*})$ only. The extension to higher dimensions is straightforward.

The LIBOR rate process $L(t, T_1^*)$ has to be a martingale under $\mathbb{P}_{T^*}$. This is achieved through the specification of (b_s) . Therefore, we choose the drift characteristic (b_s) such that

$$\int_0^t \lambda(s, T_1^*) b_s ds = -\frac{1}{2} \int_0^t c_s \lambda^2(s, T_1^*) ds - \int_0^t \int_{\mathbb{R}} \left(e^{\lambda(s, T_1^*)x} - 1 - \lambda(s, T_1^*)x \right) \nu^{T^*, L}(ds, dx). \quad (4.3)$$

With this specification $L(t, T_1^*)$ can be written as a stochastic exponential $L(t, T_1^*) = L(0, T_1^*) \mathcal{E}(H(t, T_1^*))$ where

$$H(t, T_1^*) = \int_0^t \lambda(s, T_1^*) c_s^{1/2} dW_s^{T^*} + \int_0^t \int_{\mathbb{R}} \left(e^{\lambda(s, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(ds, dx). \quad (4.4)$$

Expressed as a differential we get the $\mathbb{P}_{T^*}$ -dynamics

$$dL(t, T_1^*) = L(t-, T_1^*) \left(\lambda(t, T_1^*) c_t^{1/2} dW_t^{T^*} + \int_{\mathbb{R}} \left(e^{\lambda(t, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(dt, dx) \right) \quad (4.5)$$

with initial condition

$$L(0, T_1^*) = \frac{1}{\delta} \left(\frac{B(0, T_1^*)}{B(0, T^*)} - 1 \right).$$

Recalling that $F_B(t, T_1^*, T^*) = 1 + \delta L(t, T_1^*)$ we derive from (4.5) the dynamics of the corresponding forward process

$$\begin{aligned} dF_B(t, T_1^*, T^*) &= \delta dL(t, T_1^*) \\ &= F_B(t-, T_1^*, T^*) \left(\frac{\delta L(t-, T_1^*)}{1 + \delta L(t-, T_1^*)} \lambda(t, T_1^*) c_t^{1/2} dW_t^{T^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \frac{\delta L(t-, T_1^*)}{1 + \delta L(t-, T_1^*)} \left(e^{\lambda(t, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(dt, dx) \right). \end{aligned}$$

With the notation

$$\ell(t-, T_1^*) = \delta L(t-, T_1^*) / (1 + \delta L(t-, T_1^*)) \quad (4.6)$$

this becomes

$$\begin{aligned} dF_B(t, T_1^*, T^*) &= F_B(t-, T_1^*, T^*) \left(\ell(t-, T_1^*) \lambda(t, T_1^*) c_t^{1/2} dW_t^{T^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \ell(t-, T_1^*) \left(e^{\lambda(t, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(dt, dx) \right) \end{aligned} \quad (4.7)$$

which we write on the right-hand side as

$$F_B(t-, T_1^*, T^*) \left(\alpha(t, T_1^*, T^*) c_t^{1/2} dW_t^{T^*} + \int_{\mathbb{R}} (\beta(t, x, T_1^*, T^*) - 1) (\mu^L - \nu^{T^*, L})(dt, dx) \right)$$

for short. Based on this dynamics we define the forward martingale measure associated with date T_1^* by setting its Radon-Nikodym density as the stochastic exponential at time T_1^* ,

$$\frac{d\mathbb{P}_{T_1^*}}{d\mathbb{P}_{T^*}} = \mathcal{E}_{T_1^*}(M^1),$$

where

$$M_t^1 = \int_0^t \alpha(s, T_1^*, T^*) c_s^{1/2} dW_s^{T^*} + \int_0^t \int_{\mathbb{R}} (\beta(s, x, T_1^*, T^*) - 1) (\mu^L - \nu^{T^*, L})(ds, dx). \quad (4.8)$$

Then $W_t^{T_1^*} = W_t^{T^*} - \int_0^t \alpha(s, T_1^*, T^*) c_s^{1/2} ds$ is the forward Brownian motion for the date T_1^* and $\nu^{T_1^*, L}(dt, dx) = \beta(t, x, T_1^*, T^*) \nu^{T^*, L}(dt, dx)$ is the $\mathbb{P}_{T_1^*}$ -compensator of μ^L . Now we define the next forward LIBOR rate $L(t, T_2^*)$ by postulating that under $\mathbb{P}_{T_1^*}$

$$L(t, T_2^*) = L(0, T_2^*) \exp \left(\int_0^t \lambda(s, T_2^*) dL_s^{T_1^*} \right) \quad (4.9)$$

where

$$L_t^{T_1^*} = \int_0^t b_s^{T_1^*} ds + \int_0^t c_s^{1/2} dW_s^{T_1^*} + \int_0^t \int_{\mathbb{R}} x (\mu^L - \nu^{T_1^*, L})(ds, dx). \quad (4.10)$$

In order to make $L(t, T_2^*)$ a $\mathbb{P}_{T_1^*}$ -martingale we choose the drift term $(b_s^{T_1^*})$ appropriately, namely such that

$$\begin{aligned} \int_0^t \lambda(s, T_2^*) b_s^{T_1^*} ds &= -\frac{1}{2} \int_0^t c_s \lambda^2(s, T_2^*) ds \\ &\quad - \int_0^t \int_{\mathbb{R}} \left(e^{\lambda(s, T_2^*) x} - 1 - \lambda(s, T_2^*) x \right) \nu^{T_1^*, L}(ds, dx). \end{aligned}$$

With this specification of the drift term, $L(t, T_2^*)$ represents a stochastic exponential of the form $L(t, T_2^*) = L(0, T_2^*) \mathcal{E}(H(t, T_2^*))$ for

$$\begin{aligned} H(t, T_2^*) &= \int_0^t \lambda(s, T_2^*) c_s^{1/2} dW_s^{T_1^*} \\ &\quad + \int_0^t \int_{\mathbb{R}} \left(e^{\lambda(s, T_2^*) x} - 1 \right) (\mu^L - \nu^{T_1^*, L})(ds, dx). \end{aligned} \quad (4.11)$$

In other words, the $\mathbb{P}_{T_1^*}$ -dynamics is given by

$$\begin{aligned} dL(t, T_2^*) &= L(t-, T_2^*) \left(\lambda(t, T_2^*) c_t^{1/2} dW_t^{T_1^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \left(e^{\lambda(t, T_2^*)x} - 1 \right) (\mu^L - \nu^{T_1^*, L})(dt, dx) \right) \end{aligned} \quad (4.12)$$

with initial condition

$$L(0, T_2^*) = \frac{1}{\delta} \left(\frac{B(0, T_2^*)}{B(0, T_1^*)} - 1 \right).$$

From (4.12) we can derive the dynamics of the forward process

$$\begin{aligned} dF_B(t, T_2^*, T_1^*) &= \delta dL(t, T_2^*) \\ &= F_B(t-, T_2^*, T_1^*) \left(\frac{\delta L(t-, T_2^*)}{1 + \delta L(t-, T_2^*)} \lambda(t, T_2^*) c_t^{1/2} dW_t^{T_1^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \frac{\delta L(t-, T_2^*)}{1 + \delta L(t-, T_2^*)} \left(e^{\lambda(t, T_2^*)x} - 1 \right) (\mu^L - \nu^{T_1^*, L})(dt, dx) \right). \end{aligned}$$

As in the first step, we write $\ell(t-, T_2^*) = \delta L(t-, T_2^*) / (1 + \delta L(t-, T_2^*))$ and get

$$\begin{aligned} dF_B(t, T_2^*, T_1^*) &= F_B(t-, T_2^*, T_1^*) \left(\ell(t-, T_2^*) \lambda(t, T_2^*) c_t^{1/2} dW_t^{T_1^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \ell(t-, T_2^*) \left(e^{\lambda(t, T_2^*)x} - 1 \right) (\mu^L - \nu^{T_1^*, L})(dt, dx) \right). \end{aligned}$$

The right-hand side may be written as

$$\begin{aligned} &F_B(t-, T_2^*, T_1^*) \\ &\quad \left(\alpha(t, T_2^*, T_1^*) c_t^{1/2} dW_t^{T_1^*} + \int_{\mathbb{R}} (\beta(t, x, T_2^*, T_1^*) - 1) (\mu^L - \nu^{T_1^*, L})(dt, dx) \right). \end{aligned}$$

We can now define the forward martingale measure associated with the date T_2^* by setting

$$\frac{d\mathbb{P}_{T_2^*}}{d\mathbb{P}_{T_1^*}} = \mathcal{E}_{T_2^*}(M^2)$$

where

$$\begin{aligned} M_t^2 &= \int_0^t \alpha(s, T_2^*, T_1^*) c_s^{1/2} dW_s^{T_1^*} \\ &\quad + \int_0^t \int_{\mathbb{R}} (\beta(s, x, T_2^*, T_1^*) - 1) (\mu^L - \nu^{T_1^*, L})(ds, dx). \end{aligned} \quad (4.13)$$

Then $W_t^{T_2^*} = W_t^{T_1^*} - \int_0^t \alpha(s, T_2^*, T_1^*) c_s^{1/2} ds$ is the forward Brownian motion for the date T_2^* and

$$\begin{aligned} \nu^{T_2^*, L}(dt, dx) &= \beta(t, x, T_2^*, T_1^*) \nu^{T_1^*, L}(dt, dx) \\ &= \left(\ell(t-, T_2^*) \left(e^{\lambda(t, T_2^*)x} - 1 \right) + 1 \right) \left(\ell(t-, T_1^*) \left(e^{\lambda(t, T_1^*)x} - 1 \right) + 1 \right) F_t(dx) dt \end{aligned}$$

is the $\mathbb{P}_{T_2^*}$ -compensator of μ^L . Continuing this way we get for each time point T_j^* in the tenor structure a LIBOR rate process which can be written under the forward martingale measure $\mathbb{P}_{T_{j-1}^*}$ in the form

$$L(t, T_j^*) = L(0, T_j^*) \exp \left(\int_0^t \lambda(s, T_j^*) dL_s^{T_{j-1}^*} \right). \quad (4.14)$$

By assumption L^{T^*} , the process driving the most distant LIBOR rate $L(t, T_1^*)$, is a non-homogeneous Lévy process. This is no longer true for the processes $L^{T_j^*}$ constructed during the backward induction, since the random terms $\ell(t-, T_j^*)$ – first defined in (4.6) – enter into the compensators $\nu^{T_j^*, L}$. It simplifies the numerical implementation, which will be discussed in the next section, if all driving processes $L^{T_j^*}$ are non-homogeneous Lévy processes. This can be achieved if we approximate $\ell(t-, T_j^*)$ by the deterministic initial value $\ell(0, T_j^*) = \delta L(0, T_j^*) / (1 + \delta L(0, T_j^*))$. Then all compensators $\nu^{T_j^*, L}$ are deterministic. As pointed out by Schlögl (2002), this approximation can easily be justified, since the variation of $F_B(t, T_j^*, T_{j-1}^*)$ is several orders of magnitude smaller than the variation of $L(t, T_j^*)$. A similar approximation has been used already in the context of the swaption formula based on the lognormal forward LIBOR model (see Brace et al. 1997 and Brace and Womersley 2000). Another possibility, which produces excellent results in implementations, is to base the backward induction on the forward process. This more direct approach can be implemented without any approximation, but negative LIBOR rates can occur. We start by postulating instead of (4.1)

$$1 + \delta L(t, T_1^*) = (1 + \delta L(0, T_1^*)) \exp \left(\int_0^t \lambda(s, T_1^*) dL_s^{T^*} \right). \quad (4.15)$$

This can be written as

$$F_B(t, T_1^*, T^*) = F_B(0, T_1^*, T^*) \exp \left(\int_0^t \lambda(s, T_1^*) dL_s^{T^*} \right).$$

Choosing the drift term (b_s) as in (4.3) we can express this as a differential

$$\begin{aligned} dF_B(t, T_1^*, T^*) &= F_B(t-, T_1^*, T^*) \left(\lambda(t, T_1^*) c_t^{1/2} dW_t^{T^*} \right. \\ &\quad \left. + \int_{\mathbb{R}} \left(e^{\lambda(t, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(dt, dx) \right) \end{aligned} \quad (4.16)$$

with initial condition

$$F(0, T_1^*, T^*) = \frac{B(0, T_1^*)}{B(0, T^*)}.$$

Now we define

$$\frac{d\mathbb{P}_{T_1^*}}{d\mathbb{P}_{T^*}} = \mathcal{E}_{T_1^*}(\tilde{M}^1)$$

where

$$\tilde{M}_t^1 = \int_0^t \lambda(s, T_1^*) c_s^{1/2} dW_s^{T^*} + \int_0^t \int_{\mathbb{R}} \left(e^{\lambda(s, T_1^*)x} - 1 \right) (\mu^L - \nu^{T^*, L})(ds, dx).$$

Then $W_t^{T_1^*} = W_t^{T^*} - \int_0^t \lambda(s, T_1^*) c_s^{1/2} ds$ is the forward Brownian motion for the date T_1^* and $\nu^{T_1^*, L}(dt, dx) = \exp(\lambda(t, T_1^*)x) \nu^{T^*, L}(dt, dx)$ is the $\mathbb{P}_{T_1^*}$ -compensator of μ^L . Continuing this way we get for each T_j^* the LIBOR rate process under $\mathbb{P}_{T_{j-1}^*}$ in the form

$$1 + \delta L(t, T_j^*) = (1 + \delta L(0, T_j^*)) \exp \left(\int_0^t \lambda(s, T_j^*) dL_s^{T_{j-1}^*} \right)$$

where the successive compensators are given by

$$\nu^{T_j^*, L}(dt, dx) = \exp \left(\sum_{i=1}^j \lambda(t, T_i^*) x \right) F_t(dx) dt.$$

5 Pricing of caps and floors

An interest rate cap is a contractual agreement where the buyer of the cap places a maximum value, the strike rate, on the relevant interest rate, usually the LIBOR. He has to pay a premium to the seller of the cap to get this insurance. The premium will depend on the tenor structure $0 \leq T_0 < T_1 < \dots < T_n$, the strike rate $\mathcal{K}$, the day-count convention, and the notional amount N . Similar to swaps an interest rate cap can be settled in advance or in arrears. The time- T_j payoff of a forward cap which starts at time T_0 and is settled in arrears at dates T_j is

$$N \delta (L(T_{j-1}, T_{j-1}) - \mathcal{K})^+.$$

The counterpart to interest rate caps are interest rate floors in the same way as a call option is the counterpart of a put option. The payment of an interest rate floor settled in arrears at time T_j is

$$N \delta (\mathcal{K} - L(T_{j-1}, T_{j-1}))^+.$$

Without loss of generality we may assume that the notional amount N is one. The value at time $t \leq T_0$ is then given by

$$\text{FC}_t = \sum_{j=1}^n \mathbb{E}_{\mathbb{P}^*} \left[\frac{B_t}{B_{T_j}} \delta (L(T_{j-1}, T_{j-1}) - \mathcal{K})^+ \middle| \mathcal{F}_t \right].$$

The conditional expectation with respect to the measure $\mathbb{P}^*$ can be transformed to the conditional expectation with respect to the forward measure $\mathbb{P}_{T_j}$ by using the abstract Bayes rule (see e.g. Musiela and Rutkowski 1998, Lemma A.0.4). The density of $\mathbb{P}^*$ with respect to $\mathbb{P}_{T_j}$ is given by (see Eq. (2.13))

$$\frac{d\mathbb{P}^*}{d\mathbb{P}_{T_j}} = B_{T_j} B(0, T_j) \quad \text{and} \quad \mathbb{E}_{\mathbb{P}_{T_j}} [B_{T_j} B(0, T_j) | \mathcal{F}_t] = \frac{B_t B(0, T_j)}{B(t, T_j)}.$$

Hence

$$\begin{aligned} \text{FC}_t &= \sum_{j=1}^n \frac{B(t, T_j) \mathbb{E}_{\mathbb{P}_{T_j}} \left[(B_t/B_{T_j}) \delta(L(T_{j-1}, T_{j-1}) - \mathcal{K})^+ B_{T_j} B(0, T_j) \middle| \mathcal{F}_t \right]}{B_t B(0, T_j)} \\ &= \sum_{j=1}^n B(t, T_j) \mathbb{E}_{\mathbb{P}_{T_j}} \left[\delta(L(T_{j-1}, T_{j-1}) - \mathcal{K})^+ \middle| \mathcal{F}_t \right]. \end{aligned}$$

Each forward cap (floor) is a strip of caplets (floorlets). The j -th caplet with strike $\mathcal{K}$ and nominal value 1 is equivalent to a put option with maturity T_{j-1} , nominal value $\tilde{\delta} := 1 + \mathcal{K}\delta$, and strike $\frac{1}{\tilde{\delta}}$ on the bond with maturity T_j . The j -th caplet has positive value on the set $\{L(T_{j-1}, T_{j-1}) - \mathcal{K} > 0\}$. This is the same set where the put option described above is exercised, because

$$\begin{aligned} L(T_{j-1}, T_{j-1}) - \mathcal{K} > 0 &\iff 1 + L(T_{j-1}, T_{j-1})\delta = \frac{1}{B(T_{j-1}, T_j)} > \tilde{\delta} = 1 + \mathcal{K}\delta \\ &\iff B(T_{j-1}, T_j) < \frac{1}{\tilde{\delta}}. \end{aligned}$$

In case that the payoff of the caplet is positive, we have

$$B(T_{j-1}, T_j) \delta (L(T_{j-1}, T_{j-1}) - \mathcal{K}) = 1 - \tilde{\delta} B(T_{j-1}, T_j) = \tilde{\delta} \left(\frac{1}{\tilde{\delta}} - B(T_{j-1}, T_j) \right).$$

In the remaining part of this section we show how caplets can be priced based on an approach which uses bilateral Laplace transforms. The approach is due to Raible (2000, Chapter 3). It will be generalized and adapted for our purposes here. With a view towards implementations where purely discontinuous Lévy motions such as piecewise generalised hyperbolic Lévy processes are used as driving processes, we restrict our attention to this case. This is again to simplify notation. Thus we consider LIBOR rates $L(t, T_{j-1})$ whose dynamics under the forward measure associated with the date T_j is given by (see (4.12) – the alternative approach starting with (4.15) needs some minor modifications)

$$dL(t, T_{j-1}) = L(t-, T_{j-1}) \int_{\mathbb{R}} \left(e^{\lambda(t, T_{j-1})x} - 1 \right) (\mu^L - \nu^{T_j, L}) (dt, dx),$$

where in order to keep $\nu^{T_j, L}$ non-random we use the approximation $\ell(0, T_{j-1})$ for $\ell(t-, T_{j-1})$ as described in the previous section. Consequently

$$\begin{aligned} L(T_{j-1}, T_{j-1}) &= L(0, T_{j-1}) \exp \left(\int_0^{T_{j-1}} \lambda(s, T_{j-1}) dL_s^{T_j} \right) \\ &= L(0, T_{j-1}) \exp \left(\int_0^{T_{j-1}} b_s^{T_j} \lambda(s, T_{j-1}) ds \right. \\ &\quad \left. + \int_0^{T_{j-1}} \int_{\mathbb{R}} (x \lambda(s, T_{j-1})) (\mu^L - \nu^{T_j, L}) (ds, dx) \right). \end{aligned}$$

We define $X_t := \int_0^t \lambda(s, T_{j-1}) dL_s^{T_j}$, so that

$$X_{T_{j-1}} = \int_0^{T_{j-1}} b_s^{T_j} \lambda(s, T_{j-1}) ds + \int_0^{T_{j-1}} \int_{\mathbb{R}} x \lambda(s, T_{j-1}) (\mu^L - \nu^{T_j, L})(ds, dx).$$

In the same way as in Eberlein and Raible (1999, Lemma 3.1 and Formula (3.4)) we get the characteristic function χ of $X_{T_{j-1}}$ in the following form:

$$\begin{aligned} \chi(u) &:= \mathbb{E}_{\mathbb{P}_{T_j}} \left[e^{iu X_{T_{j-1}}} \right] = \exp \left(iu \int_0^{T_{j-1}} b_s^{T_j} \lambda(s, T_{j-1}) ds \right. \\ &\quad \left. + \int_0^{T_{j-1}} \int_{\mathbb{R}} \left(e^{iu \lambda(s, T_{j-1}) x} - 1 - iu \lambda(s, T_{j-1}) x \right) \nu^{T_j, L}(ds, dx) \right). \end{aligned}$$

With the choice of the drift characteristic $(b_s^{T_j})$ made in Sect. 4 this becomes

$$\begin{aligned} \chi(u) &= \exp \left(-iu \int_0^{T_{j-1}} \int_{\mathbb{R}} \left(e^{\lambda(s, T_{j-1}) x} - 1 - \lambda(s, T_{j-1}) x \right) \nu^{T_j, L}(ds, dx) \right) \quad (5.1) \\ &\quad + \int_0^{T_{j-1}} \int_{\mathbb{R}} \left(e^{iu \lambda(s, T_{j-1}) x} - 1 - iu \lambda(s, T_{j-1}) x \right) \nu^{T_j, L}(ds, dx) \\ &= \exp \left(\int_0^{T_{j-1}} \int_{\mathbb{R}} \left(e^{iu \lambda(s, T_{j-1}) x} - iu e^{\lambda(s, T_{j-1}) x} - (1 - iu) \right) \nu^{T_j, L}(ds, dx) \right). \quad (5.2) \end{aligned}$$

According to our model

$$(L(t, T_{j-1}))_{t \in [0, T_{j-1}]} = (L(0, T_{j-1}) \exp(X_t))_{t \in [0, T_{j-1}]}$$

is a martingale with respect to the forward measure associated with the date T_j . Hence the moment generating function of $X_{T_{j-1}}$

$$\text{mgf}(u) := \mathbb{E}_{\mathbb{P}_{T_j}} \left[e^{u X_{T_{j-1}}} \right]$$

exists at least for $u \in [0, 1]$ and we have

$$\text{mgf}(1) = \mathbb{E}_{\mathbb{P}_{T_j}} \left[e^{X_{T_{j-1}}} \right] = \mathbb{E}_{\mathbb{P}_{T_j}} \left[\frac{L(T_{j-1}, T_{j-1})}{L(0, T_{j-1})} \right] = 1.$$

We make the following technical assumptions, cf. Raible (2000, Sects. 3.1 and 3.2). We assume that $\int_{-\infty}^{\infty} |\chi(u)| du < \infty$, so that the distribution of $X_{T_{j-1}}$ has a Lebesgue density, i.e. there exists a Lebesgue-measurable function ρ such that

$$\frac{d\mathbb{P}_{T_j}^{X_{T_{j-1}}}}{d\text{Leb}} = \rho, \quad \text{Leb-a.s.}$$

We define $w(x, \mathcal{K}) := (x - \mathcal{K})^+$ so that the payoff of the j -th caplet is $\delta w(L(T_{j-1}, T_{j-1}), \mathcal{K})$. The price of the j -th caplet at time 0 is given by

$B(0, T_j) \delta \mathbb{E}_{\mathbb{P}_{T_j}} [w(L(T_{j-1}, T_{j-1}), \mathcal{K})]$. We introduce the auxiliary function $v_{\mathcal{K}}$ by $v_{\mathcal{K}}(x) := w(e^{-x}, \mathcal{K})$.

Note that for $R < -1$ the function $x \mapsto e^{-Rx} v_{\mathcal{K}}(x)$ is bounded since

$$0 \leq e^{-Rx} (e^{-x} - \mathcal{K})^+ = \left(e^{(|R|-1)x} - \mathcal{K} e^{|R|x} \right)^+ \leq \frac{1}{|R|} \left(\frac{|R| - 1}{\mathcal{K}|R|} \right)^{|R|-1}.$$

The function $x \mapsto e^{-Rx} v_{\mathcal{K}}(x)$ is strictly greater than zero if and only if $x < -\ln(\mathcal{K})$. Hence $x \mapsto e^{-Rx} v_{\mathcal{K}}(x)$ is integrable because

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-Rx} (e^{-x} - \mathcal{K})^+ dx &= \int_{-\infty}^{-\ln(\mathcal{K})} \left(e^{(|R|-1)x} - \mathcal{K} e^{|R|x} \right) dx \\ &= \frac{\mathcal{K}^{1-|R|}}{|R|^2 - |R|} < \infty. \end{aligned}$$

Theorem 5.1 *Let $\zeta_j := -\ln(L(0, T_{j-1}))$. Then $L(T_{j-1}, T_{j-1}) = e^{-\zeta_j + X_{T_{j-1}}}$. Assume that $\text{mgf}(-R) < \infty$. Let $V_j(\zeta_j, \mathcal{K})$ be the time-0 price of the j -th caplet and let $L[v_{\mathcal{K}}]$ be the bilateral Laplace transform of $v_{\mathcal{K}}$, i.e.*

$$L[v_{\mathcal{K}}](z) = \int_{-\infty}^{\infty} e^{-zx} v_{\mathcal{K}}(x) dx, \quad z = R + iu \in \mathbb{C}, u \in \mathbb{R}.$$

Then

$$V_j(\zeta_j, \mathcal{K}) = \delta B(0, T_j) \frac{e^{\zeta_j R}}{2\pi} \lim_{M \rightarrow \infty} \int_{-M}^M e^{iu\zeta_j} L[v_{\mathcal{K}}](R + iu) \chi(iR - u) du \quad (5.3)$$

whenever the right-hand side exists.

Proof The price of the j -th caplet at time 0 can be written in the following way:

$$\begin{aligned} V_j(\zeta_j, \mathcal{K}) &= \delta B(0, T_j) \mathbb{E}_{\mathbb{P}_{T_j}} [w(L(T_{j-1}, T_{j-1}), \mathcal{K})] \\ &= \delta B(0, T_j) \mathbb{E}_{\mathbb{P}_{T_j}} [v_{\mathcal{K}}(\zeta_j - X_{T_{j-1}})] \\ &= \delta B(0, T_j) \int_{\mathbb{R}} v_{\mathcal{K}}(\zeta_j - x) \mathbb{P}_{T_j}^{X_{T_{j-1}}} (dx) \\ &= \delta B(0, T_j) \int_{\mathbb{R}} v_{\mathcal{K}}(\zeta_j - x) \rho(x) dx \\ &= \delta B(0, T_j) (v_{\mathcal{K}} * \rho)(\zeta_j). \end{aligned}$$

Thus,

$$L[V_j](R + iu) = \delta B(0, T_j) L[v_{\mathcal{K}}](R + iu) L[\rho](R + iu) \quad \text{for } u \in \mathbb{R}. \quad (5.4)$$

The integral that defines the bilateral Laplace transform $L[V_j](z)$ converges absolutely and $\zeta \mapsto V_j(\zeta, \mathcal{K})$ is a continuous function by Theorem B.2 in Raible (2000).

Hence the payoff V_j can be obtained by inverting the Laplace transform, see Raible (2000, Theorem B.3). So

$$\begin{aligned} V_j(\zeta_j, \mathcal{K}) &= \frac{1}{2\pi i} \int_{R-i\infty}^{R+i\infty} e^{\zeta_j z} L[V_j](z) dz \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{\zeta_j(R+iu)} L[V_j](R+iu) du \\ &= \frac{e^{\zeta_j R}}{2\pi} \lim_{M \rightarrow \infty} \int_{-M}^M e^{iu\zeta_j} L[V_j](R+iu) du. \end{aligned}$$

By Eq. (5.4) we get

$$V_j(\zeta_j, \mathcal{K}) = \delta B(0, T_j) \frac{e^{\zeta_j R}}{2\pi} \lim_{M \rightarrow \infty} \int_{-M}^M e^{iu\zeta_j} L[v_{\mathcal{K}}](R+iu) \chi(iR-u) du. \quad \square$$

It is sufficient to consider the case where the strike price equals one, since by Lemma 3.3 in Raible (2000),

$$V_j(\zeta_j, \mathcal{K}) = \mathcal{K} V_j(\zeta_j + \ln \mathcal{K}, 1).$$

The bilateral Laplace transform $L[v_{\mathcal{K}}]$ for $\mathcal{K} = 1$ is given by $L[v_1](z) = (z(z+1))^{-1}$. The method of using bilateral Laplace transforms is also suitable for more complicated payoff functions as long as the payoff depends only on X_{T_j-1} . Examples are given in Raible (2000, Table 3.1).

Let us finally remark that a topic of future research will be the modeling of defaultable LIBOR rates, see Lotz and Schlögl (2000), Schönbucher (2000), and Bielecki and Rutkowski (2002). In analogy to (2.7), the defaultable δ -(forward)-LIBOR rates are defined by

$$L^d(t, T) := \frac{1}{\delta} \left(\frac{D_C(t, T)}{D_C(t, T + \delta)} - 1 \right),$$

where $D_C(t, T)$ denotes the time- t price of the defaultable bond with maturity T which can be modeled according to the approach of Eberlein and Özkan (2003).

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